Deterministic Optimization

Discrete Optimization:

Linear Programming Relaxations

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Linear Programming Relaxation

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Learning Objectives

- Examine the information provided by the LP relaxation of an integer program

LP Relaxation

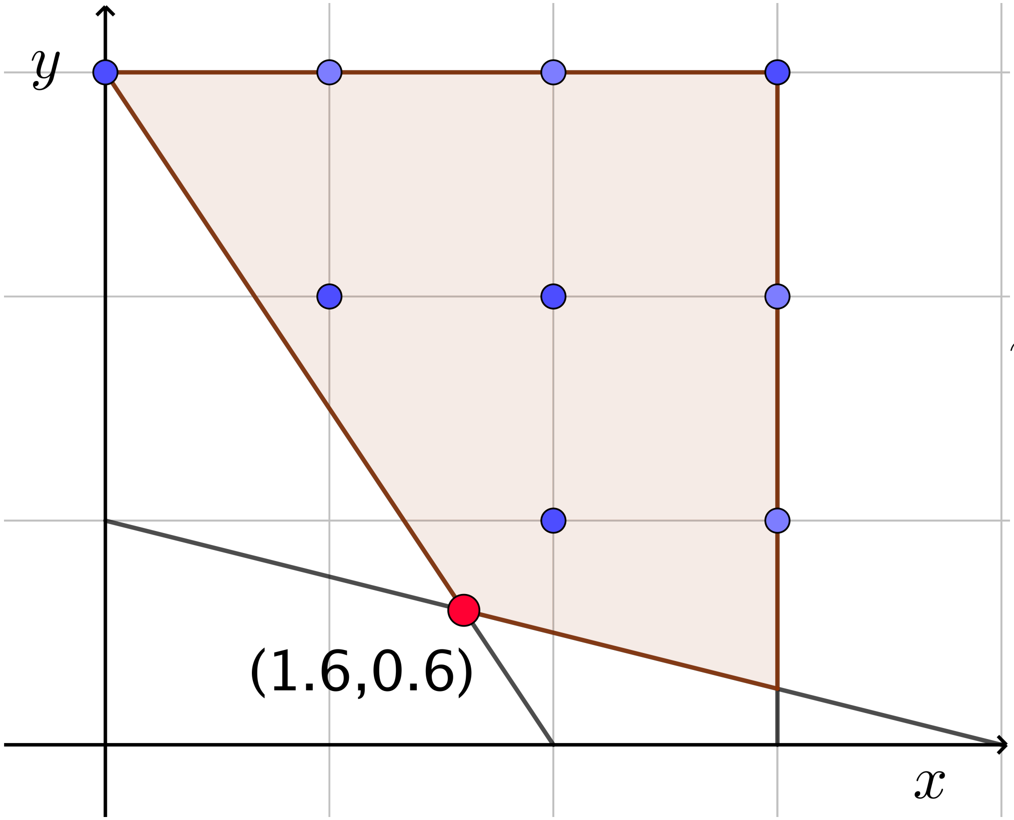
Consider a mixed-integer linear program (MILP):

$$\begin{aligned} v_{IP} = \min \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \geq \mathbf{b} \\ & \mathbf{x} \in \mathbb{R}^{n-p} \times \mathbb{Z}^p \end{aligned}$$

It's linear programming relaxation is:

$$\begin{aligned} v_{LP} = \min \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} \geq \mathbf{b} \\ & \mathbf{x} \in \mathbb{R}^{n} \end{aligned}$$

Example



$$
\begin{aligned}
v_{IP} = \min \quad & x_1 + y_1 \\
\text{s.t.} \quad & 3x + 2y \geq 6 \\
& x + 4y \geq 4 \\
& 0 \leq x \leq 3,\ 0 \leq y \leq 3 \\
& x,\ y \text{ integer}
\end{aligned}
$$

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LP Relaxation Properties

- If the LP relaxation is infeasible then so is the IP.
- (As long as all data is rational) If the LP relaxation is unbounded then the IP can be either infeasible or unbounded.
- If the LP relaxation has an optimal solution, then the IP could be infeasible or have an optimal solution.
- It always holds that $v_{LP} \leq v_{IP}$ (for minimization problems)
- If an optimal solution to the LP relaxation is integral, then it is an optimal solution to the IP.

Optimality Gap

- The optimal value of the LP relaxation allows us to estimate the optimal gap of a given feasible solution to the IP.

- Example: Consider the solution $(2,1)$ which is feasible and has an objective value of 3. The LP relaxation has an objective value of 2.2. Thus $2.2 \leq v_{IP} \leq 3$. The optimality gap of the solution $(2,1)$ is at most $(3-2.2) = 0.8$.

$$
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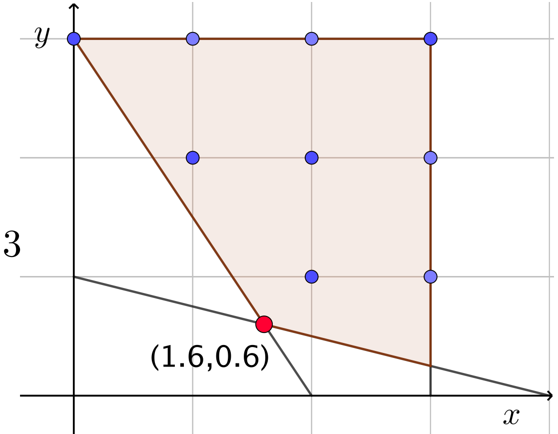
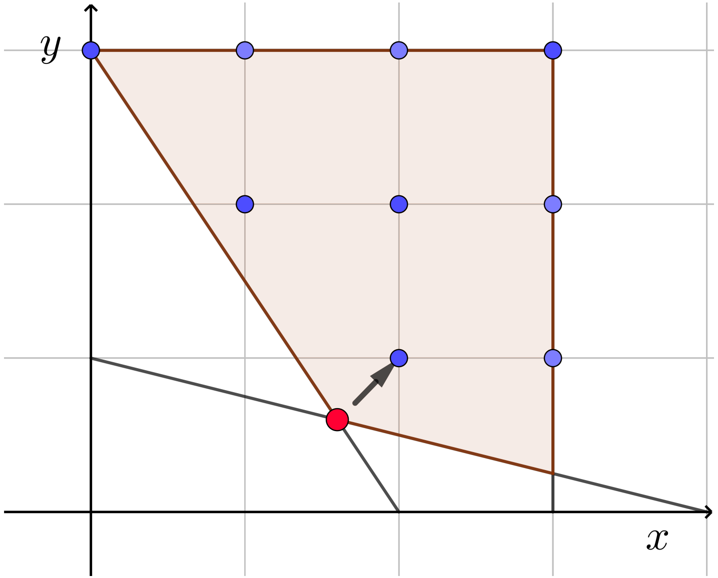
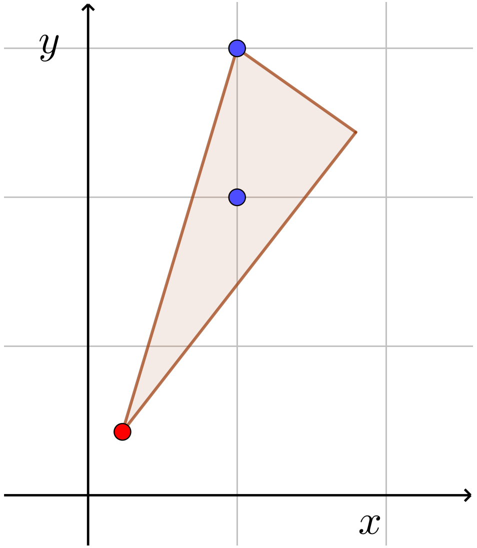


Rounding LP Solution

- Sometimes by rounding the LP relaxation solution we may be able to find a good solution to the IP.





Summary

- The LP relaxation provides valuable information about an IP
- It’s optimal value provides bounds and optimality gap estimates
- It’s solutions can sometimes be rounded to get a good solution to the IP